

Research Article

A Novel Numerical Treatment of Nonlinear and Nonequilibrium Model of Gradient Elution Chromatography considering Core-Shell Particles in the Column

Abdulaziz Garba Ahmad ¹, Mohammed K. A. Kaabar ^{2,3}, Saima Rashid ⁴, and Muhammad Abid⁵

¹Department of Mathematics Programme, National Mathematical Centre, Abuja, Nigeria

²Gofa Camp, Near Gofa Industrial College and German Adebabay, Nifas Silk-Lafto 26649, Addis Ababa, Ethiopia

³Institute of Mathematical Sciences, Faculty of Science, University of Malaya, Kuala Lumpur 50603, Malaysia

⁴Department of Mathematics, Government College University, Faisalabad 38000, Pakistan

⁵Department of Mathematics, COMSATS University Islamabad, Islamabad, Pakistan

Correspondence should be addressed to Mohammed K. A. Kaabar; mohammed.kaabar@wsu.edu

Received 6 February 2022; Accepted 21 April 2022; Published 13 May 2022

Academic Editor: Nagarajan Deivanayagampillai

Copyright © 2022 Abdulaziz Garba Ahmad et al. This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

An extended method of semidiscrete high-resolution finite volume is used in this paper to obtain numerical solutions for a formulated nonlinear lumped kinetic model of liquid chromatographic process to examine the effect of chromatographic column overloading gradient elution considering core-shell particles. The model constitutes linear solvent strength (LSS), Henry's constant, coefficient of nonlinearity, and coefficient of axial dispersion. The effects of modulator concentration changes for the elution of single and two components are analyzed. The advantages of introducing gradient elution against isocratic elution in terms of core radius fraction are investigated intensively. Numerical temporal moments are generated from the solutions obtained for a more in-depth examination of the considered model. Moreover, multiple forms of a single- and two-component mixture are generated to analyze the influences of core radius fractions on gradient elution. For example, the obtained results are utilized to investigate the effects of the slope of gradient, concentration of modulator, solvent strength parameter, coefficient of nonlinearity, coefficient of mass transfer, and coefficient of axial dispersion on the profiles of concentration in order to improve the process performance using optimal core radius fraction parameter values.

1. Introduction

Chromatography is a laboratory technique that has become a widespread and popular tool for separating and purifying the mixture into its components. This technique can be utilized to separate chemical compounds that other traditional methods such as distillation, filtration, or extraction cannot in chemical analysis. It is used in chemical engineering and forensic science and has many other diverse industrial applications. Also, this method is used by environmental protection agencies to test the purity of drinking water and the quality of the air [1–3].

Gradient elution in liquid chromatography is a process that is generated by changing the configuration of the mobile phase over time while keeping the flow rate constant during the gradient run [4]. So, the composition of the liquid phase remains fixed (constant) throughout the process of high-performance liquid chromatographic (HPLC) separation, and the operation is referred to as an isocratic elution. Gradient elution chromatography can be used to separate molecules with widespread shifting elution peaks [5, 6]. This chromatography method was introduced in the 1950s and had a wide range of analytical and preparation applications. In this type of chromatography, the strength of elution can

be increased or decreased by changing the mobile phase composition. The technique significantly improves analysis time, peak resolution, and peak detection capabilities, which has piqued the interest among researchers in gradient elution chromatography [7]. Because the retention behavior of the solvent in this operating state changes with the composition of the mobile phase, theoretical investigation for gradient elution appears to be more complicated than those for isocratic elution. Snyder [8, 9] is acknowledged for contributing significantly to gradient elution theory by formulating an extensively utilized concept of linear solvent strength (LSS) gradients. The gradient program is modified in an LSS-gradient approach so that the logarithms of the sample compounds' retention factors decrease linearly [10]. Snyder and Dolan presented a comprehensive investigation of the linearized solvent strength gradient approach [11]. LSS gradients are employed for predicting results in the chromatographic reverse phase, where the gradient elution technique is widely used. Non-LSS highly curved plots of retention factor over modulator concentration, on the other hand, have been implemented in various chromatographic methods, specifically in ion-exchange chromatography.

In order to increase the effectiveness of HPLC columns, different packing materials have been investigated [12–15]. The utilization of core-shell particles has increasingly captivated the interest of many researchers in both the analytical and preparatory liquid chromatography separation processes [16–18]. These were used to separate peptides as well as other molecules like nucleotides and proteins [12, 16]. Aside from that, some theoretical findings on the use of core-shell particles were made by analyzing one-dimensional (1D) liquid chromatography models. Kaczmarek and Guichon [19] have published a detailed analysis of the general rate model, which was used to prove the benefits of the thin-shell coated beads. Also, Wang et al. [15] investigated the relationship between the pressure flow ion-exchange resin and the core-shell particles. Further, Luo et al. [20] analyzed that when compared to completely porous homogeneous agarose beads, core beads increased vital stiffness and permeability. The most completed model, that is, the general rate model (GRM), was used to investigate and optimize the effects on multicomponent isocratic elution by using the core radius fraction by Gu et al. [21]. Qamar et al. and his fellow researchers have analyzed the effectiveness of fully porous particles and also some with the core-shell particles by developing and solving several chromatographic separation models [18, 22].

To understand the chromatographic separation processes, a number of mathematical models existing in the literature might be employed [3, 23, 24]. Each model has its own complexity level in describing the chromatography processes. The intricacies of these existing mathematical models are based on the suppositions made about the behavior of the thermodynamic system, adsorption-desorption kinetics, kinds of packing particles used, and also the dispersion and mass transfer mechanisms. The general rate model (GRM), the lumped kinetic model, and the dispersive equilibrium model are the most frequently and commonly utilized models in liquid chromatography. In the literature,

multiple numerical techniques like finite volumes method, finite-difference method, and finite element methods have been employed to simulate the chromatographic models [25–27].

In this work, a one-dimensional nonlinear lumped kinetic model of gradient elution chromatography packed with core-shell particles in the column is formulated and numerically approximated in order to theoretically investigate the effect of core radius fraction in column overloading gradient elution. In the developed model, Henry's constants parameter, coefficients of nonlinearity, coefficients of mass transfer, and the coefficients of axial dispersion rely on the composition of the solvent. The present study extends our previous analysis [28], which is based on nonlinear (1D) LKM of gradient elution chromatography considering fully porous particles. Contrary to the previous analysis, this current research considered core-shell particles loaded in the column. Analytical solutions of the model equations for nonlinear adsorption isotherm are not possible to acquire. Numerical simulations are required for this purpose to precisely predict the behavior of a chromatographic column dynamically. Several techniques have been discussed in the literature to obtain the approximate solution of nonlinear partial differential equations [29]. To solve the governing equations arising in the given model, the high-resolution finite volume method (HR-FVM) is developed and implemented. As it is illustrated in our last research article, this method allows second-to-third-order accuracy in the solution. Furthermore, numerical temporal moments are derived from the derived solutions for a more in-depth examination of the considered model. The implications of different parameters on process performance are investigated, as well as the advantages of gradient elution instead of isocratic elution in the simulation analysis. The obtained results also demonstrate the model's and numerical scheme's strength and potential in situations more similar to real-world applications.

The remainder of this presentation is divided into several parts. In Section 2, we formulate the model's specification and description of nonlinear 1D-LKM incorporated with core-shell particles and overloading gradient elution in the chromatographic column. In Section 3, we present the numerical simulation of the nonisothermal 1D-LKM with the considered Dirichlet boundary conditions. In Section 4, we describe the approximate results obtained. In the last section, we summarize the concluding remarks.

2. Model Configuration and Formulation

A nonlinear 1D-LKM of gradient elution chromatography is formulated for an isothermal column filled with core-shell particles as packing material. The process of adsorption in the column is controlled through axial dispersions and the resistance to mass transfer phenomena. The LKM uses a driving force that is linear in the solid phase and only investigates one extra parameter to supplement the coefficient of axial dispersion. Also, the coefficient of mass transfer combines the effects of mass transport resistances both internally and externally. The model equations are

derived based on the following assumptions, which are considered in the model formulations. The mobile phase is perfectly incompressible. Furthermore, the cored beads have a uniform radius R_p and a core radius fraction $\zeta_{core} = R_c/R_p$, where R_c is the radius of the inert core. The total mass transfer rate in a column is connected with two kinetic parameters, the mass transfer rate coefficient that is represented by K_L , and the coefficient of axial

dispersion that is symbolized by D_z . This lumped kinetic model 1D-LKM is employed to study the dynamic behavior in the chromatographic separation process considering core-shell particles and overloading gradient elution in the column. Considering the above assumptions, the mass balance and kinetic equations for the movement of N_c species in the mobile phase are given as [28, 30]

$$\frac{\partial u_i}{\partial t} + v \frac{\partial u_i}{\partial z} = D_{z,i}(\phi) \frac{\partial^2 u_i}{\partial z^2} - F(1 - \zeta_{core})K_L(\theta)(\phi_i^* - \phi_i), \quad i = 1, 2, \dots, N_c. \quad (1)$$

The specific number in the mixture sample is indicated by N_c in the equation above. For every component of i , u_i , ϕ_i , and ϕ_i^* denote the solute concentration, the solute concentration in the immobilize phase, and the concentration of solute adsorbed by the stationary phase, respectively, in the bulk of the fluid. K_L represents the coefficient of external mass transfer, v represents the interstitial velocity, t denotes the time, z denotes the axial coordinate, and $F = (1 - \epsilon/\epsilon)$ is the phase ratio, while ϵ is the external porosity.

The diffusion and adsorption-desorption processes in the stagnant mobile phase are assumed to be infinitely fast in this model. As a result, in order to complete the model, the following basic linear driving force method is applied to assess the accumulation in the solid phase:

$$\frac{\partial \phi_i}{\partial t} = F(1 - \zeta_{core})K_L(q)(\phi_i^* - \phi_i), \quad i = 1, 2, \dots, N_c. \quad (2)$$

An isotherm that is symbolized as ϕ_i^* describes the relationship between the equilibrium adsorbate concentrations in the liquid phase and the equilibrium adsorption quantities in the solid phase. The isotherms can be used to simulate the equilibrium adsorption data. Also, it is employed by researchers to examine adsorption information such as adsorption mechanisms, adsorbent properties, and maximum adsorption capacity. A variety of adsorption isotherm models, including linear, Langmuir, and Freundlich isotherms, have been established in the literature. In the following study, the adsorption equilibrium was modelled using the multicomponent Langmuir equation [3]. The composition of the mobile phase varies throughout the gradient elution process due to changes in a modifier concentration. The following is a formulation of the local equilibrium:

$$\phi_i^*(u_i, q) = \frac{K_{H,i}(q)u_i}{1 + \sum_{j=1}^{N_c} b_j(q)u_j}. \quad (3)$$

Henry's coefficient is represented by $K_{H,i}$ for the i -th component. The degree of nonlinearity, which is connected with the isotherm for the i -th number of components in the mixture, is quantified by $b_{r,i}$. In most cases, experimentally determining the functional dependencies of the two isotherm parameters on c is required. These two strategies are simultaneously utilized to model the retention of analytes as

a function of the organic solvent fraction. The mathematical description of the LSS model is

$$b_i(q) = b_{r,i}e^{-\alpha q}, \quad D_{z,i}(q) = D_{zr,i}e^{-\alpha q}, \quad K_{L,i}(q) = K_{Lr,i}e^{-\alpha q}, \quad \& K_{H,i}(q) = K_{Hr,i}e^{-\alpha q}. \quad (4)$$

Here, $K_{Hr,i}$, $K_{Lr,i}$, $b_{r,i}^{ref}$, and $D_{zr,i}$ represent the reference values of the mass transfer coefficient, Henry's constant, nonlinearity, and axial dispersion coefficient. The specific solvent strength is symbolized by q , and α is the volume-fraction parameter of the updating nonretained solvent.

It turns out that determining the distribution of the mobile phase's strong solvent across the column is required before investigating solute retention behavior during elution because it calculates the solute's migration velocity at any time and as well as at any point within the column. Based on the preceding assumptions, the optimal model for calculating the change in the volume fraction of the strong solvent parameter in the mobile phase is [1]

$$\frac{\partial q}{\partial t} + v \frac{\partial q}{\partial z} = 0. \quad (5)$$

Here, we defined the initial and boundary conditions for the model equations:

$$q(z, 0) = q_0, \quad 0 \leq z \leq \mathcal{L}, \quad (6)$$

$$q(0, t) = \begin{cases} q_0, & \text{if } t < t_s, \\ q(t - t_s), & \text{if } t_s \leq t \leq t_e, \\ q_e, & \text{if } t > t_e. \end{cases} \quad (7)$$

For a linear gradient approach, the following applies to the case:

$$q(t) = q_0 + \beta t, \quad \beta = \frac{q_e - q_0}{t_e - t_s}. \quad (8)$$

In light of the above, the term β is the slope (or steepness) of the gradient, q_0 and q_e are the initial and final volume fraction of the modifier (in the mobile phase composition), q represents the implemented profile of the gradient, and t_s and t_e are the starting and ending time of the gradient.

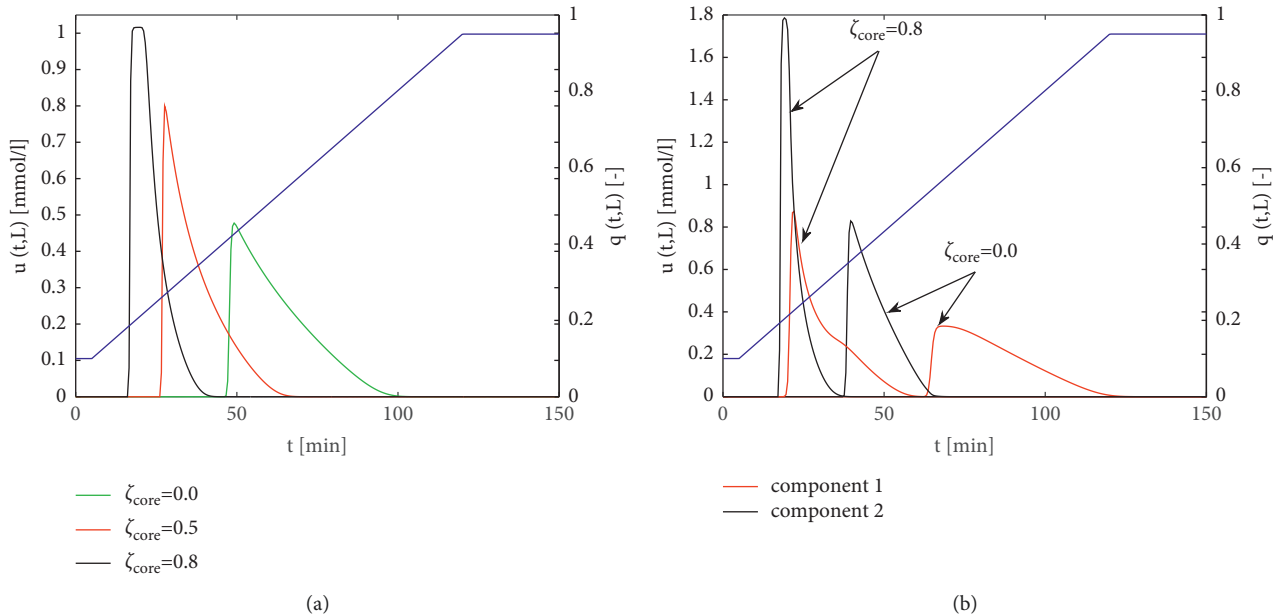
The initial and boundary conditions are added to complete the gradient elution model for lumped kinetic considering core-shell particles in the column:

TABLE 1: Used parameters for linear solvent strength case for single component.

Parameters name	Numeric values
Length of chromatographic column	$\mathcal{L} = 10 \text{ cm}$
Porosity	$\varepsilon = 0.4$
Interstitial velocity	$v = 1.0 \text{ cm/min}$
Reference mass transfer coefficient	$k_{Lr} = 10 \text{ min}^{-1}$
Reference Henry's coefficient	$k_{Hr} = 7$
Reference coefficient of axial dispersion	$D_{zr} = 0.002 \text{ cm}^2/\text{min}$
Reference nonlinearity coefficient	$b^{bref} = 2.0$
Initial concentration	$q_0 = 0.1$
Final concentration	$q_e = 0.95$
Gradient starting time	$t_s = 5 \text{ min}$
Gradient ending time	$t_e = 120 \text{ min}$
Solvent strength parameter	$\alpha = 0.90$

TABLE 2: Used parameters for linear solvent strength case for two components.

Parameters name	Numeric values
Length of chromatographic column	$\mathcal{L} = 10 \text{ cm}$
Interstitial velocity	$v = 0.6 \text{ cm/min}$
Porosity	$\varepsilon = 0.4$
Reference coefficient of mass transfer	$k_{Lr} = 10 \text{ min}^{-1}$
Reference coefficient of axial dispersion	$D_{zr} = 0.0002 \text{ cm}^2/\text{min}$
Reference Henry's constant for component-I	$k_{Hr,1} = 10$
Reference Henry's constant for component-II	$k_{Hr,2} = 15$
Injected concentration	$u_{inj,1} = 1.0 \text{ mol/l}$
Injected concentration	$u_{inj,2} = 1.0 \text{ mol/l}$
Injection time	$t_{inj} = 10.0 \text{ min}$
Solvent strength parameter	$\alpha_1 = 0.90$
Solvent strength parameter	$\alpha_2 = 0.40$
Solvent strength parameter	$\gamma_1 = 0.90$
Solvent strength parameter	$\gamma_2 = 0.40$
Reference nonlinearity coefficient for component-I	$b_1^{bref} = 1$
Reference nonlinearity coefficient for component-II	$b_2^{bref} = 2$
Gradient starting time	$t_s = 5 \text{ min}$
Gradient ending time	$t_e = 120 \text{ min}$
Initial concentration	$q_0 = 0.10$
Final concentration	$q_e = 0.95$

FIGURE 1: The elution profile of single- and two-component for distinct values of core radius fraction ζ_{core} . Table 1 lists all of the other parameter values that were chosen.

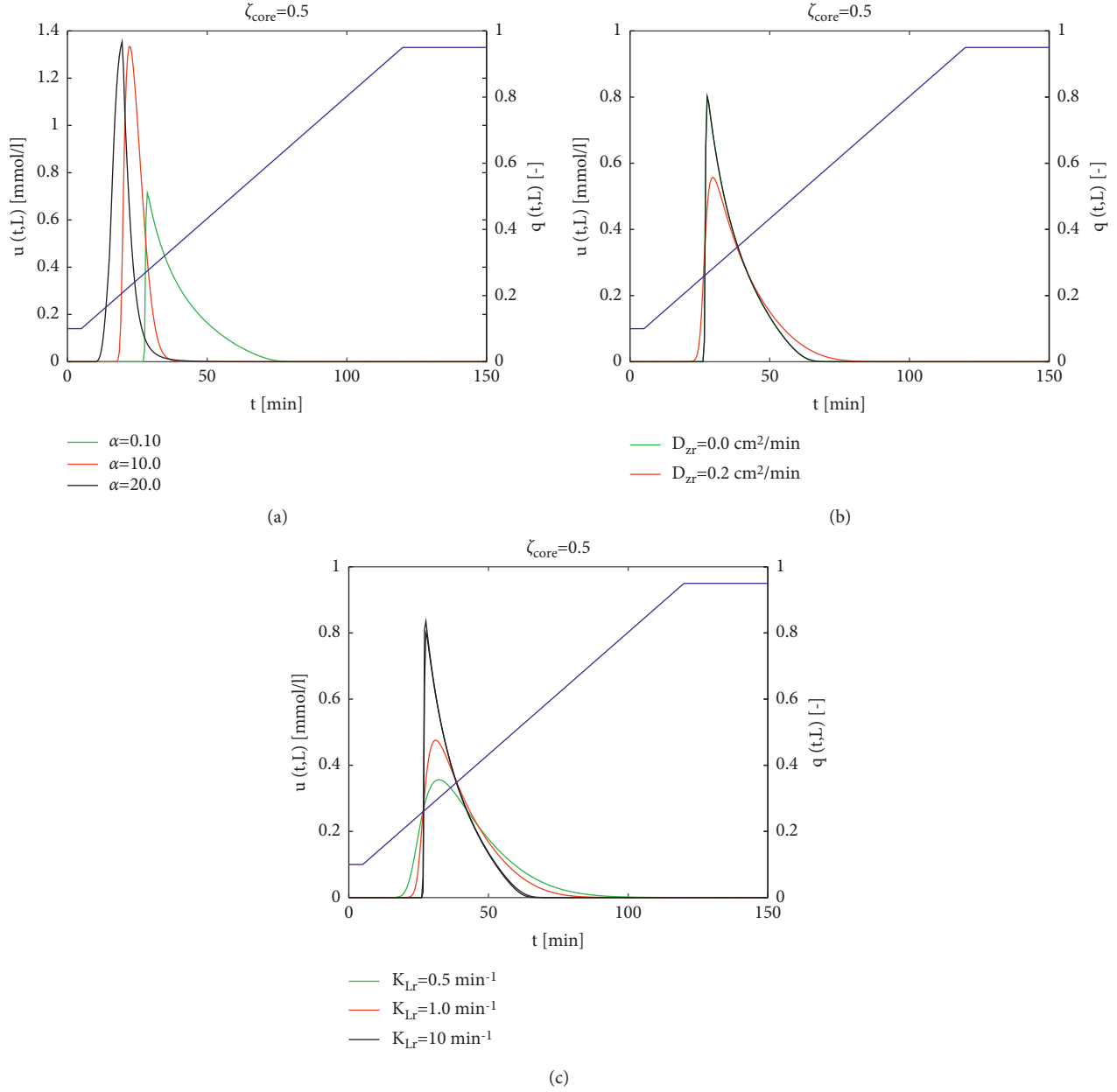


FIGURE 2: Here, the value of $\zeta_{core} = 0.5$ is applied to check the effects of (a) parameter of specific component α , (b) reference coefficient of nonlinearity b^{ref} , (c) reference coefficient of axial dispersion D_{zr} , and (d) reference coefficient external mass transfer K_{Lr} on nonlinear one-dimensional single-component elution profiles. Table 1 lists all of the other parameter values that were chosen.

$$u_i(z, 0) = 0, \phi_i(z, 0) = 0, i = 1, 2, 3, \dots, N_c, \quad (9)$$

$$u_i(0, t) = \begin{cases} u_{i, inj}, & \text{if } 0 \leq t \leq t_{inj}, \\ 0, & t > t_{inj}, \end{cases} \quad i = 1, 2, 3, \dots, N_c. \quad (10)$$

These inlet BCs work with the following Neumann condition at the right end of the chromatographic column:

$$\frac{\partial u_i}{\partial z}(L, t) = 0, i = 1, 2, \dots, N_c, \quad (11)$$

where the injected concentration for the i -th component is indicated by the symbol $u_{i, inj}$ and the dimensionless time of

injection is denoted by t_{inj} . Moreover, the injection becomes continuous when the time of the simulation process is less than the injection time t_{inj} or when $t_{inj} \rightarrow \infty$.

3. Numerical Method

Numerous numerical procedures for approximating chromatographic models are known in literature [1, 3, 25]. The model equations are solved in this section using a semi-discrete high-resolution flux-limiting finite volume procedure. Moreover, this procedure was recently employed on gradient elution chromatography models [28, 31].

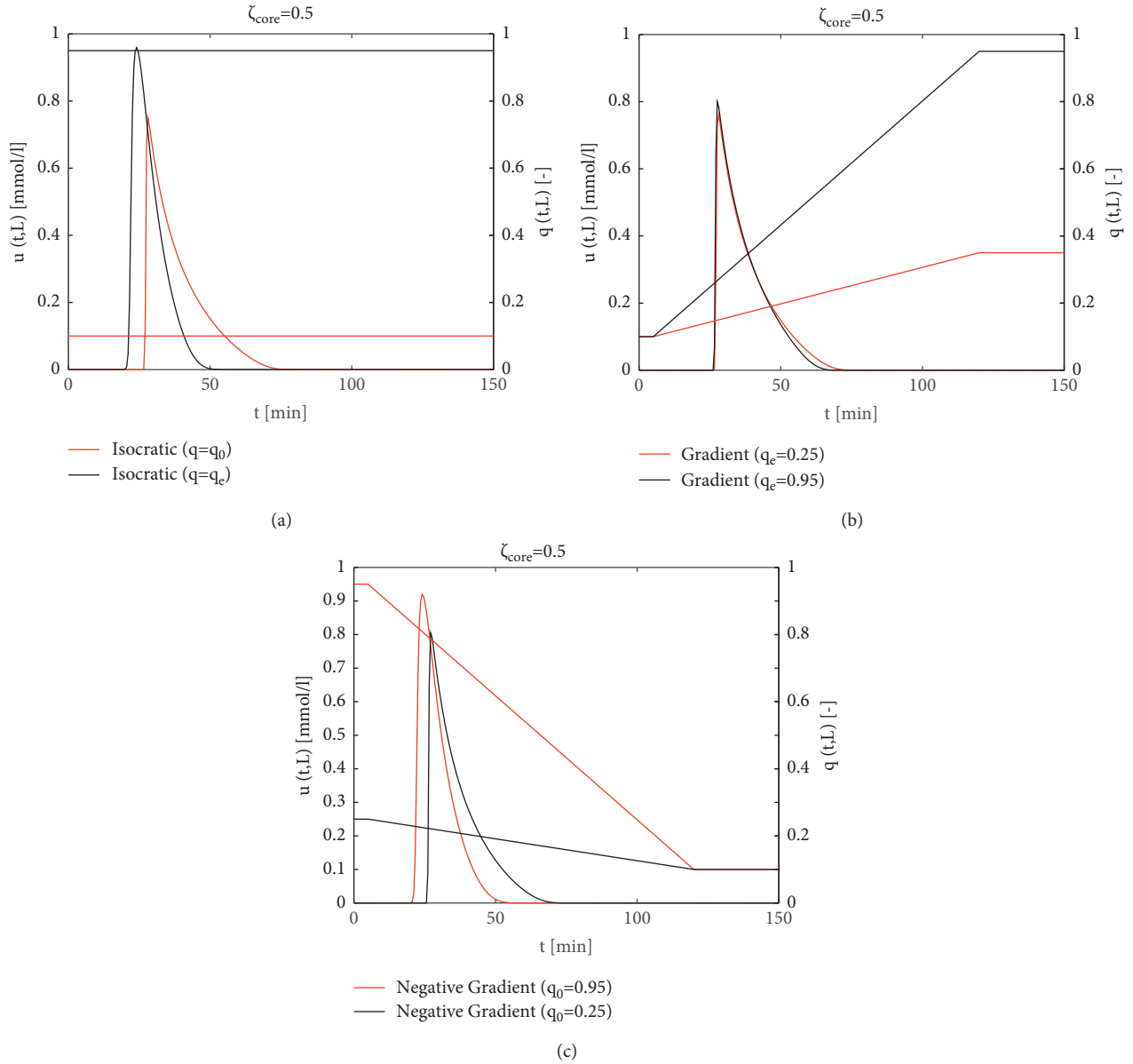


FIGURE 3: Here, in this case, the parameter $\zeta_{core} = 0.5$ is used to show the comparison of elution profiles for isocratic and gradient, including positive and negative profiles of the gradients utilizing the elution for a single component. Table 1 contains a list of other parameters.

3.1. Domain Discretization. The first stage in executing the scheme into practice is to discretize the computing domain. The primary and essential goal of the following discretization procedure is to generate a set of time-coordinated coupled ODEs.

The mesh interval width is constant and represented by $\Delta\xi$; let $\mathcal{N}_\xi$ be the large integer representing the number of grid points. The nodal point ξ_j encircled by the cell for $1 \leq h \leq \mathcal{N}_\xi$ in the interval $\Omega_h \equiv [\xi_{h-1/2}, \xi_{h+1/2}]$ is as follows:

$$\xi_{1/2} = 0, \xi_{h+1/2} = h\Delta\xi, \xi_{\mathcal{N}+1/2} = \mathcal{L}, \quad (12)$$

$$\xi_h = \frac{\xi_{h-1/2} + \xi_{h+1/2}}{2}, \Delta\xi_h = \xi_{h+1/2} - \xi_{h-1/2} = \frac{\mathcal{L}}{\mathcal{N}_\xi + 1}. \quad (13)$$

The averaged initial data $\lambda_h(0)$ are defined in each cell as $\lambda_h(0) = \lambda_h(t) = \frac{1}{\Delta\xi_h} \int_{\Omega_h} \lambda(\xi, 0) d\xi$, $h = 1, 2, \dots, N$. (14)

Here, $\lambda \in \{u_i, \phi_i, \phi_i^*\}$.

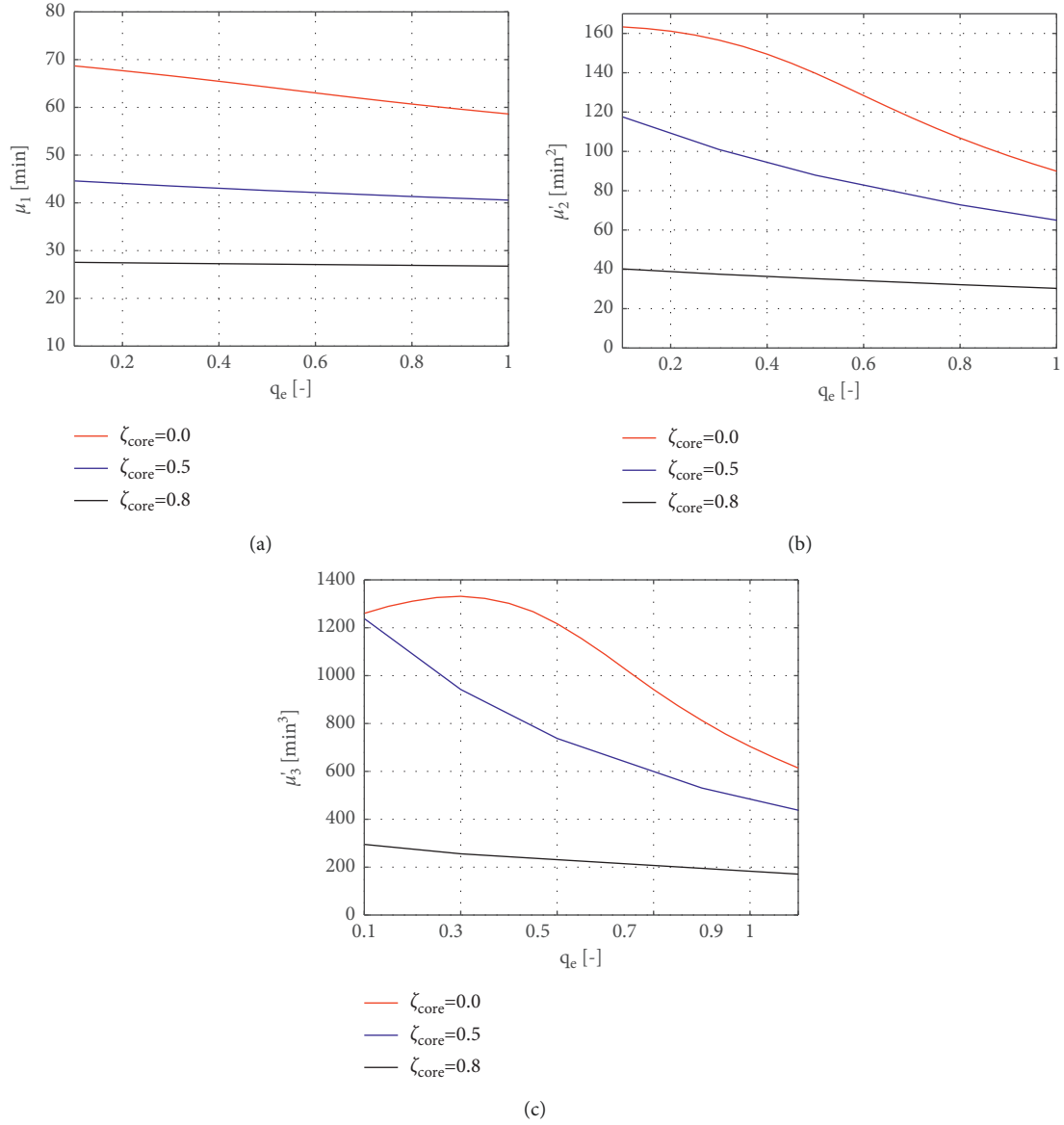


FIGURE 4: Here, three different values of ζ_{core} are used to check the effect of modulator concentration q_e on temporal profile of moments. q_e has been significantly updated, while the other parameters remain unchanged as shown in Table 1.

This can help to simplify the cell averages λ_h^{n+1} at the next stage of time, that is, t^{n+1} , if we know the cell averages at t^n . Moreover, integrating (1) and (2) over Ω_h gives

$$\frac{du_{i,h}}{dt} = -v \frac{u_{i,h+1/2} - u_{i,h-1/2}}{\Delta z} + \frac{D_{z,i}(q_h(t))}{\Delta z} \left[\left(\frac{\partial u_i}{\partial z} \right)_{i,h+1/2} - \left(\frac{\partial u_i}{\partial z} \right)_{i,h-1/2} \right] - F(1 - \zeta_{core})K_L(q_h(t))(\phi_i^* - \phi_i), \quad (15)$$

$$\frac{d\phi_{i,h}}{dt} = F(1 - \zeta_{core})K_L(q_h(t))(\phi_i^* - \phi_i), \quad i = 1, 2, \dots, N_c. \quad (16)$$

Volume fraction of the solvent $q_h(t)$ is given as

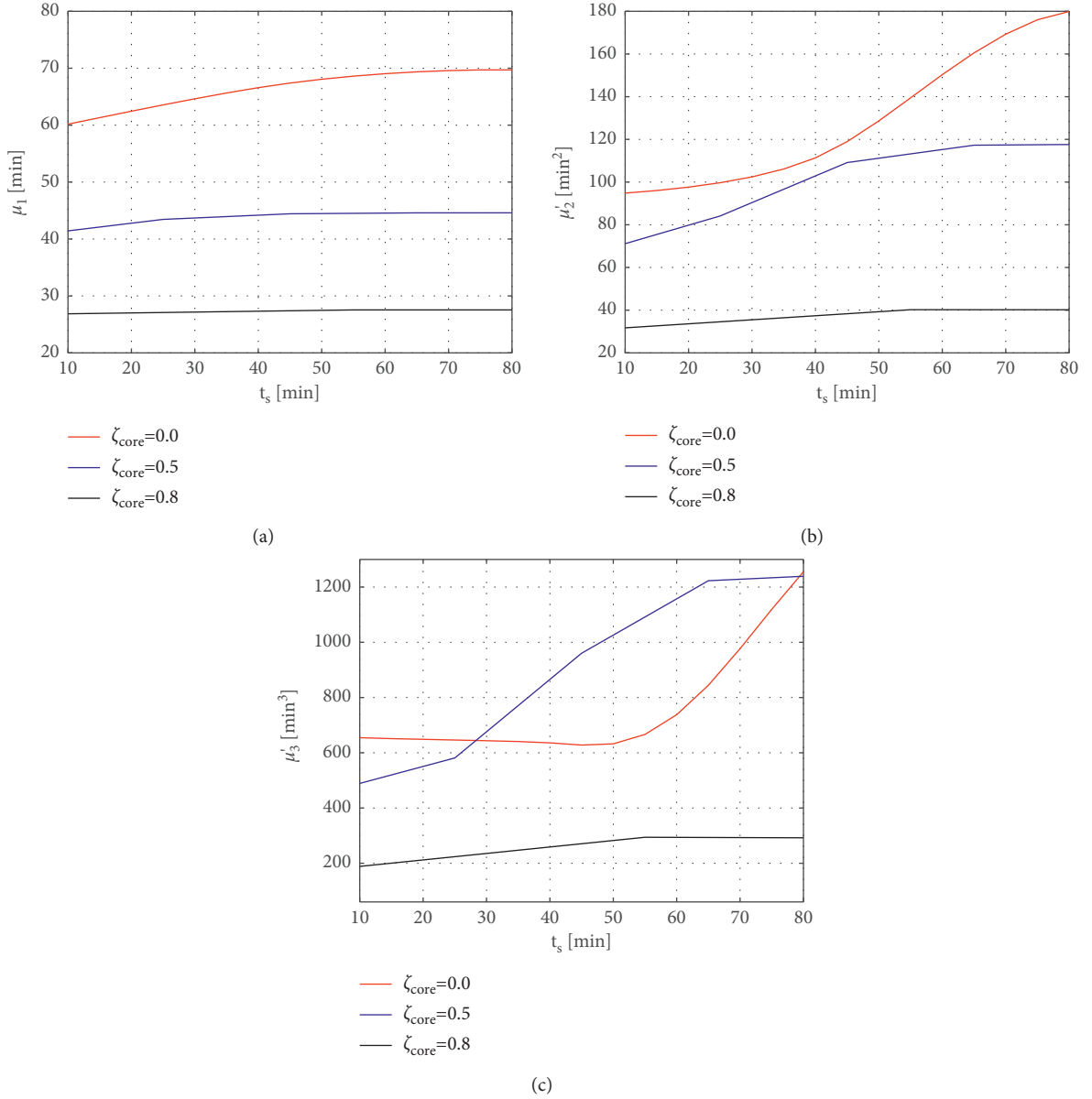


FIGURE 5: Here, three different values of ζ_{core} are used to check the effect of gradient initial time t_s on numerical temporal moments. While other parameters are fixed, t_s has been gradually modified. Table 1 displays all of the other parameter values that were chosen.

$$q_h(t) = \begin{cases} q_0, & \text{if } t - \frac{z_h}{u} < t_s, \\ q\left(t - \frac{z_h}{u} - t_s\right), & \text{if } t_s \leq t - \frac{z_h}{u} \leq t_e, \\ q_e, & \text{if } t - \frac{z_h}{u} > t_e. \end{cases} \quad (17)$$

The differential terms arise in the diffusion part of (15), which can be approximated as follows:

$$\left[\frac{\partial u_i}{\partial z} \right]_{h \pm 1/2} = \left[\frac{u_{i,h+1} - u_{i,h}}{\Delta z} \right]. \quad (18)$$

The further step is to calculate concentrations at interfaces of the cell. There are numerous techniques for approximating these fluxes, resulting in a wide range of numerical systems. The first- and second-order approximations are presented here.

3.2. First-Order Approximation. Concentration values are estimated at interfaces of the cell in (18) by employing the formula of backward difference. Also, for the term of concentration, the first-order approximation is represented as

$$u_{i,h+1/2} = u_{i,h}, \quad u_{i,h-1/2} = u_{i,h-1}. \quad (19)$$

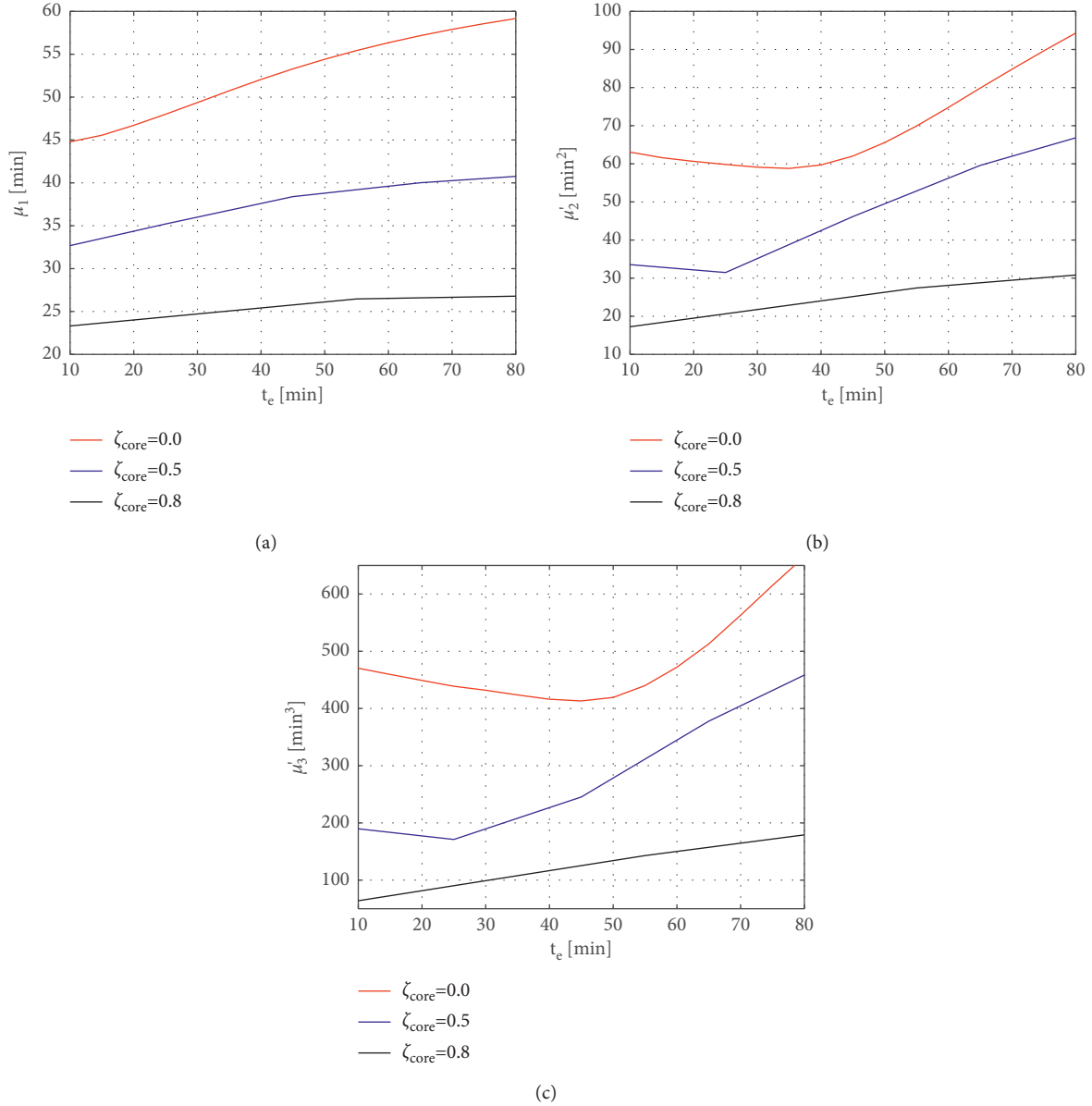


FIGURE 6: Here, three different values of ζ_{core} are used to check the effect of gradient final time t_e on the temporal profile of moments. t_e has been steadily updated, although other parameters have stayed unchanged, as seen in Table 1, which includes all of the other parameter values that were selected.

3.3. Second-Order Approximation. The flux-limiting formulas described in the following are used to approximate concentration values at the cell's interface [32]. Now, $u_{i,h+1/2}$ is defined as

$$u_{i,h+1/2} = u_{i,h} + \frac{1}{2} \Psi(r_{i,h+1/2})(u_{i,h} - u_{i-1,h}), \quad (20)$$

and

$$r_{h+1/2} = \frac{u_{i,h+1} - u_{i,h} + \eta}{u_{i,h} - u_{i,h-1} + \eta}. \quad (21)$$

Furthermore, to control division by zero, we require $\eta = 10^{-10}$. Here, Ψ (the flux-limiting functions) is utilized to

keep the numerical scheme's local monotonicity in (20), which can be described as [32]

$$\Psi[r_{h+1/2}] = \max \left[(0, \min \left[2r_{h+1/2}, \min \left[\frac{1}{3} + \frac{2}{3}r_{i+1/2}, 2 \right] \right] \right] \right]. \quad (22)$$

The flux-limiting approximations do not cover the boundary intervals. Take into account the left boundary, which also has an in-flow boundary condition. This interval faces $\xi_{1/2}$, and the in-flow boundary is in the same place. Consequently, cell-interface concentration is simulated using the first-order backward procedure. The fluxes at the boundaries of inner cells can be calculated using the previously described (second-order accurate) HR-FVM. At the

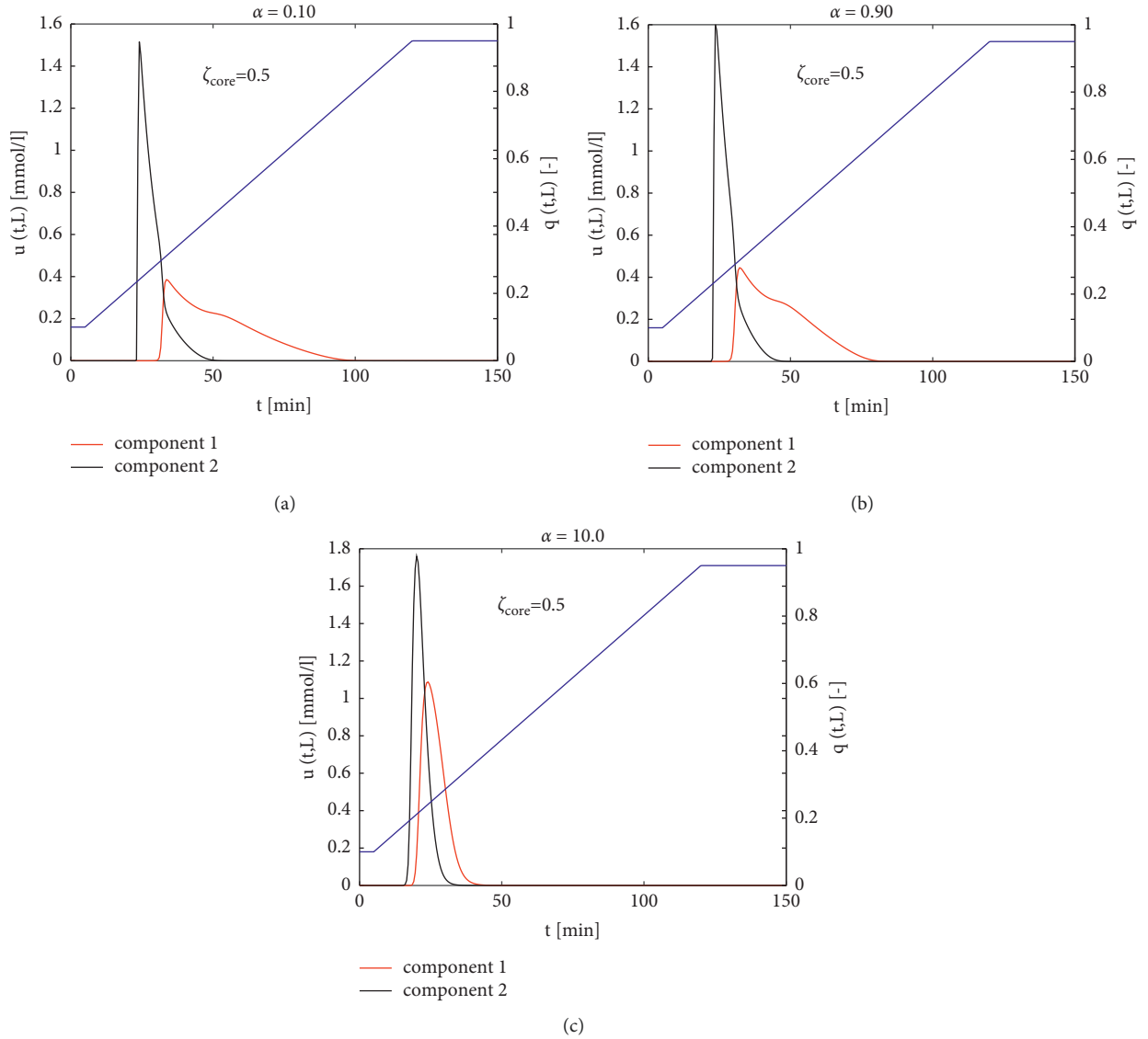


FIGURE 7: Here, the value of $\zeta_{core} = 0.5$ is employed to check the effect of α (i.e., the specific parameter of solvent strength on profiles of elution). Table 2 illustrates how α has increased steadily while other parameters have stayed unchanged.

cell boundaries $\xi_{3/2}$ and $\xi_{N+1/2}$, the first-order approximation can be utilized to solve this problem. Let $u_{i,inj}$ represent the injected flux, and then

$$u_{i,1/2} = u_{i,inj}, u_{i,1/2} = u_{i,1}, u_{i,N+1/2} = u_{i,N}. \quad (23)$$

4. Numerical Temporal Moments

Temporal profiles of the moments can be utilized to demonstrate the graphical findings obtained in the chromatogram for the duration between the moving phases of the chromatographic process. It is an effective approach for collecting fundamental informative details regarding liquid chromatography elution in a column. Chromatographic phenomena are affected by both retention equilibrium and kinetic mass transfer inside the column, for instance,

column efficiency, the profile of the elution peak, sample retention, separation performance, band broadening, and other variables. The simulated moments are generated using the numerical methods we proposed. n -th temporal moment at the outlet of the chromatographic bed of length $z = \mathcal{L}$ is confined as the n -th temporal moment to obtain the simulated numerical moments.

$$\mathcal{M}_n = \int_0^\infty (z = \mathcal{L}, t) t^n dt. \quad (24)$$

The n -th normalised temporal moment is donated as

$$\mu_n = \frac{\mathcal{M}_n}{\mathcal{M}_0} = \frac{\int_0^\infty u(z = \mathcal{L}, t) t^n dt}{\int_0^\infty u(z = \mathcal{L}, t) dt}. \quad (25)$$

The n -th central moment is given as

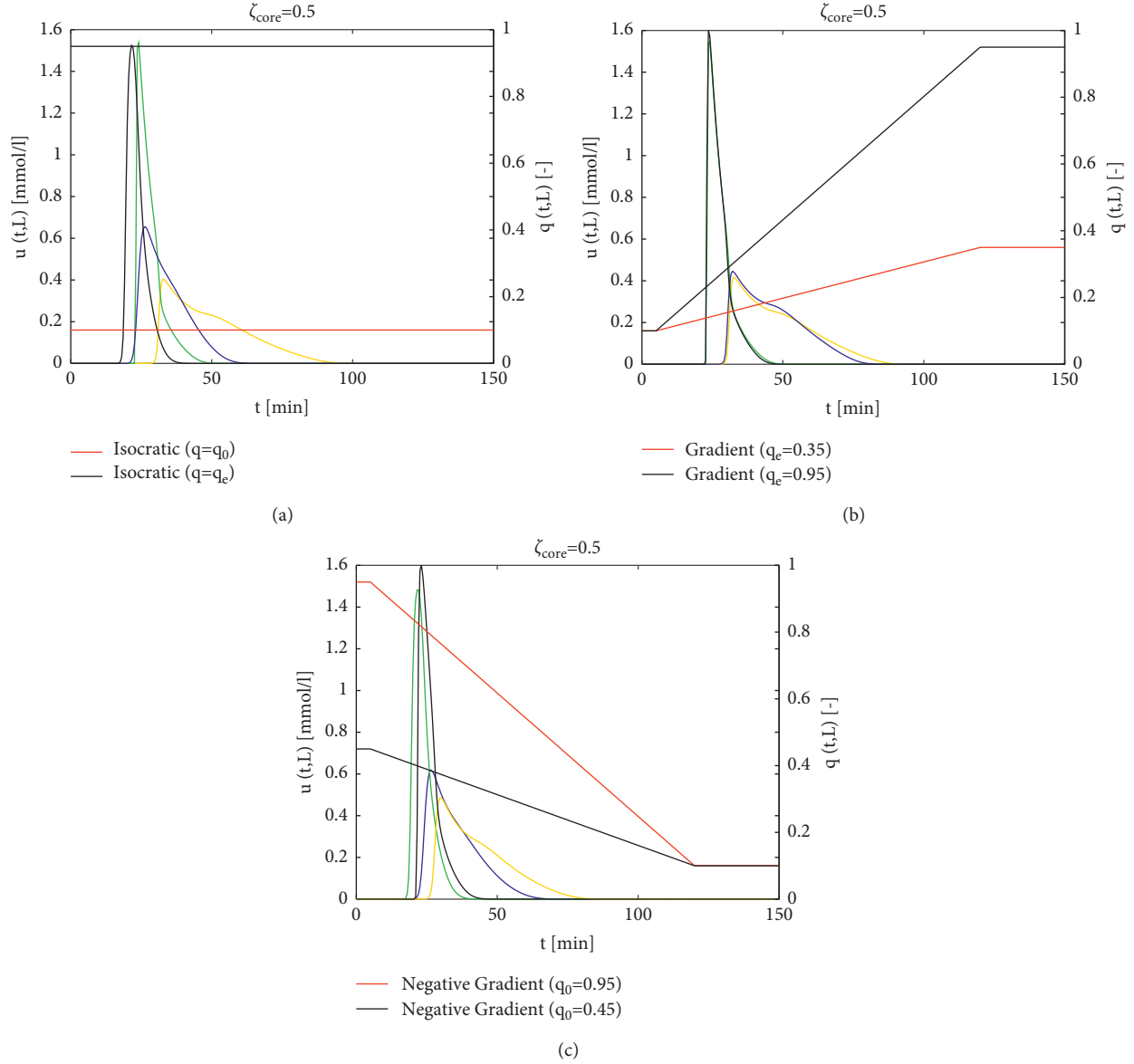


FIGURE 8: Here, the value of $\zeta_{core} = 0.5$ is employed for comparing the elution profile of isocratic and gradient for both positive and negative profiles of the gradients sequentially utilizing the elution profile for two components. Table 2 has a range of required parameters.

$$\mu_i^* = \frac{\int_0^\infty u(z = \mathcal{L}, t) (t - \mu_1)^n dt}{\int_0^\infty u(z = \mathcal{L}, t) dt}. \quad (26)$$

Continuous injections might be carried out using derivatives of the given concentration profiles ($t_{inj} \rightarrow \infty$) to transform stage responses that enable moments to be calculated [33]. The central moments used to analyze continuously successful curves are therefore presented as

$$\mu_i^* = \frac{\int_0^\infty \dot{u}(z = \mathcal{L}, t) (t - \mu_1)^n dt}{\int_0^\infty \dot{u}(z = \mathcal{L}, t) dt}, \text{ where } \dot{u} = \frac{du}{dt}. \quad (27)$$

The elution curve, also known as the probability density curve, is utilized in liquid chromatography to characterize the concentration distribution concerning time. These density curves are determined by the statistical moments,

including zeroth, first, second, and third moments. The zero moments, which are indicated by the variations in the concentration profiles, are used to determine the area of peak or total mass of the solvent. The third moment includes information regarding the peak's asymmetry or skewness.

5. Numerical Results and Discussion

The suggested HR-FVM is used in this section to simulate the impacts of various gradient elution chromatographic separation procedures. Here, in this numerical case study, the chromatographic processes of single- and two-component solutes are investigated. To facilitate the simulation process, some assumptions have been made, the Henry coefficient $K_{H,i} = k_{Hr,i}$, the mass transfer coefficients $K_{L,i} = k_{Lr}$, the axial dispersion coefficients

$D_{z,i} = D_{zr}$, and the nonlinearity coefficient $b_i = b^{\text{bref}}$. The rest of all other remaining parameters are listed in Tables 1 and 2. A nonlinear 1D LKM given by equations (1)–(4) was analyzed. The plots show q (the modulator concentration) and u (concentration of the solute) at the given outlet of the column $z = \mathcal{L}$, which are plotted against time t . However, in practice, they may be different, and present model equations account for such variations.

In Figure 1, a case is investigated for distinct values of core radius fraction ζ_{core} , for the solution profile of (a) nonlinear single-component elution profile and (b) nonlinear two-component elution profile. It can be deeply evaluated that the profiles of elution become sharper by increasing ζ_{core} from 0 to 0.8 and decrease their times of retention accordingly; that is, the efficiency of the column gradually increases by enlargement of the value of ζ_{core} on both the single- and two-component plots. Similarly, the holding capacity of the column reduces gradually.

Figure 2 demonstrates the effects of α , b^{bref} , D_{zr} , and k_{Lr} for core radius fraction value $\zeta_{\text{core}} = 0.5$. Figure 2(a) displays the impact on the strength of the solvent parameter α examined on the nonlinear single-component profile. As α changes from 0.1 to 1.0, the retention time decreases, that is, as the sensitivities of Henry's constant to significant difference increases in the modulator concentration profile. Also, the peak of the profile increases higher due to the influence of ζ_{core} from the core-shell particles. Figure 2(b) shows the influence of the nonlinearity parameter b^{bref} on the elution profile. For the increasing value of the coefficient of nonlinearity, the elution peaks become tailed, and the retention time decreases. However, for $b^{\text{bref}} = 0$, the elution profile has a representation of Gaussian shape. This describes the Langmuir behavior that is normally noticed in liquid chromatography processes. Figure 2(c) depicts the influence of the parameter of coefficient of axial dispersion D_{zr} on the elution profiles for two distinct values. D_{zr} is not affecting the mean retention while it becomes peaked for increasing the value of D_{zr} . In Figure 2(d), this shows that elution profiles become more sharpened and peaked for increasing the value of k_{Lr} . Contrarily, as expected, the mass transfer coefficient does not affect the average retention time.

In Figure 3, it demonstrates the comparison among the elution profiles for isocratic and gradient for both positive and negative gradients' profiles sequentially for the elution of a single component. Here, the value of the core radius fraction $\zeta_{\text{core}} = 0.5$ is utilized. In Figure 3(a), it can be seen that isocratic elution produces a bigger profile with more asymmetry and a longer transportation time. However, the isocratic profile with $q = q_e$ generates a narrower profile, which is symmetric. Figure 3(b) displays the elution profiles generated by the positive value of gradient elution, which are symmetric, narrower, and most importantly asymmetric. In next Figure 3(c), the impact of negative gradient elution results in a smaller and peaked elution profile.

Figure 4 presents the results of three different values of ζ_{core} on temporal moments to check the influence of gradually changing values modulator concentration q_e . For

the higher value of ζ_{core} , the profile of the temporal moment has a high magnitude which produces the gradual reduction of the holding capacity of the column. For the first moment, it described the mean retention time. Also, its value reduces with higher values of q_e . Similarly, the second and third moments can be seen with the same behavioral effect.

Figure 5 demonstrates the plots of moments to analyze the influence of the beginning time of gradient t_s for a fixed final time t_e . Here, three different values of ζ_{core} were utilized for core-shell particles. For the first moment, it described the meantime of retention. For the smaller value of ζ_{core} , its value rises with increasing values of t_s . A similar behavioral effect can be noticed for the moment plots of the second as well as the third moment, which described the variance and asymmetry of the eluent profile. In addition, due to the impact of ζ_{core} , the elution profile becomes peaked.

Figure 6 analyzes the effect of ending time t_e on the elution profile. Here, three different values of ζ_{core} were utilized for the impact of core-shell particles. It is clear from the figures that a smaller value ζ_{core} results in a larger slope of the gradient and consequently results in shortened residence time of the elution profile in the chromatographic column. For the plots of the second and third moments, a similar behavioral impact can be observed. Also, due to the impact of ζ_{core} , the elution profile become peaked.

Figure 7 presents the numerically simulated results for distinct values of the parameter of solvent strength α on the two-component elution profile. Here, the values of $\zeta_{\text{core}} = 0.5$ are used. When the value of α is increased, the time spent by the pulses inside the column decreases. Once again, due to the impact of ζ_{core} , the elution profile becomes peaked.

Figure 8 uses a two-component elution profile to analyze the eluent profiles of isocratic and gradient elution for both positive and negative gradients. Here, a similar behavioral effect can be noticed, like the one we discussed earlier in Figure 3.

6. Conclusion

In the present paper, a nonlinear 1D-LKM of gradient elution chromatography packed with core-shell particles in the column was developed and numerically approximated to analyze the effect of core radius fraction in column overloading gradient elution. The model solution was obtained numerically using an HR-FVM considering a suitable flux limiter. The nonlinear isotherm was analyzed in which the Henry's constant and the coefficients of nonlinearity both depended on the composition of the solvent. The study showed that larger values of core radius fraction generate profiles of sharper peaks and small residence times. Thus, the efficiency of the column was increased because of a reduction in the diffusion path inside the adsorbents. Additionally, the other parameters' effects on the separate components that behave differently inside the chromatographic column were also investigated. Moreover, the numerical solutions obtained illustrate that core-shell particles of suitable core radius fraction can be used for optimizing the liquid chromatography processes.

Data Availability

No data were used to support this study.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Authors' Contributions

Abdulaziz Garba Ahmad contributed to actualization, validation, methodology, formal analysis, investigation, and initial draft. Mohammed K. A. Kaabar contributed to actualization, methodology, formal analysis, validation, investigation, initial draft, and supervision of the original draft and editing. Saima Rashid contributed to actualization, methodology, formal analysis, validation, investigation, and initial draft. Muhammad Abid contributed to actualization, validation, methodology, formal analysis, investigation, and initial draft. All authors read and approved the final version.

Acknowledgments

The authors would like to express their sincere thanks to the National Mathematical Centre, Abuja, Nigeria, for supporting this study.

References

- [1] G. Guiochon, "Preparative liquid chromatography," *Journal of Chromatography A*, vol. 965, no. 1-2, pp. 129–161, 2002.
- [2] D. M. Ruthven, *Principles of Adsorption and Adsorption Processes*, John Wiley & Sons, Hoboken, New Jersey, United States, 1984.
- [3] G. Guiochon, A. Felinger, and D. G. Shirazi, *Fundamentals of Preparative and Nonlinear Chromatography*, Elsevier, Amsterdam, Netherlands, 2006.
- [4] S. Qamar, N. Rehman, G. Carta, and A. Seidel-Morgenstern, "Analysis of gradient elution chromatography using the transport model," *Chemical Engineering Science*, vol. 225, p. 115809, 2020.
- [5] J. C. Touchstone, "History of chromatography," *Journal of Liquid Chromatography*, vol. 16, no. 8, pp. 1647–1665, 1993.
- [6] M. Tswett, "Physikalisch-chemische Studien über das chlorophyll. Die Adsorptionen," *Berichte der Deutschen Botanischen Gesellschaft*, vol. 24, pp. 316–323, 1906.
- [7] L. Zechmeister, "History, scope, and methods of chromatography," *Annals of the New York Academy of Sciences*, vol. 49, no. 2, pp. 145–160, 1948.
- [8] L. R. Snyder, "Linear elution adsorption chromatography," *Journal of Chromatography A*, vol. 13, pp. 415–434, 1964.
- [9] L. R. Snyder, "Principles of gradient elution," *Chromatographic Reviews*, vol. 7, pp. 1–51, 1965.
- [10] L. R. Snyder, J. W. Dolan, and J. R. Gant, "Gradient elution in high-performance liquid chromatography," *Journal of Chromatography A*, vol. 165, no. 1, pp. 3–30, 1979.
- [11] L. R. Snyder and J. W. Dolan, *The Linear-Solvent-Strength Model of Gradient Elution. In Advances in Chromatography*, CRC Press, Boca Raton, Florida, United States, 2021.
- [12] J. J. Kirkland, F. A. Truszkowski, C. H. Dilks Jr, and G. S. Engel, "Superficially porous silica microspheres for fast high-performance liquid chromatography of macromolecules," *Journal of Chromatography A*, vol. 890, no. 1, pp. 3–13, 2000.
- [13] S. Fekete, K. Ganzler, and J. Fekete, "Facts and myths about columns packed with sub-3 μ m and sub-2 μ m particles," *Journal of Pharmaceutical and Biomedical Analysis*, vol. 51, no. 1, pp. 56–64, 2010.
- [14] Y. Xiang, B. Yan, C. V. McNeff, P. W. Carr, and M. L. Lee, "Synthesis of micron diameter polybutadiene-encapsulated non-porous zirconia particles for ultrahigh pressure liquid chromatography," *Journal of Chromatography A*, vol. 1002, no. 1-2, pp. 71–78, 2003.
- [15] X. Wang, W. E. Barber, and P. W. Carr, "A practical approach to maximizing peak capacity by using long columns packed with pellicular stationary phases for proteomic research," *Journal of Chromatography A*, vol. 1107, no. 1-2, pp. 139–151, 2006.
- [16] J. Ning, F. Kong, D. Li, and Y. Du, "Preparation of monodisperse agglomerated pellicular anion-exchange resins compatible with high-performance liquid chromatography solvents for ion chromatography," *Journal of Chromatography A*, vol. 793, no. 1, pp. 193–197, 1998.
- [17] S. A. Schuster, B. M. Wagner, B. E. Boyes, and J. J. Kirkland, "Wider pore superficially porous particles for peptide separations by HPLC," *Journal of Chromatographic Science*, vol. 48, no. 7, pp. 566–571, 2010.
- [18] A. G. Ahmad, S. Perveen, and S. Qamar, "Numerical approximation of a two-dimensional non-equilibrium model of non-isothermal liquid chromatography considering core-shell particles," *Journal of Liquid Chromatography & Related Technologies*, vol. 43, no. 17-18, pp. 793–808, 2020.
- [19] K. Kaczmarek and G. Guiochon, "Modeling of the mass-transfer kinetics in chromatographic columns packed with shell and pellicular particles," *Analytical Chemistry*, vol. 79, no. 12, pp. 4648–4656, 2007.
- [20] J. Luo, W. Zhou, Z. Su, G. Ma, and T. Gu, "Comparison of fully-porous beads and cored beads in size exclusion chromatography for protein purification," *Chemical Engineering Science*, vol. 102, pp. 99–105, 2013.
- [21] T. Gu, M. Liu, K.-S. C. Cheng, S. Ramaswamy, and C. Wang, "A general rate model approach for the optimization of the core radius fraction for multicomponent isocratic elution in preparative nonlinear liquid chromatography using cored beads," *Chemical Engineering Science*, vol. 66, no. 15, pp. 3531–3539, 2011.
- [22] S. Qamar, S. Bashir, S. Perveen, and A. Seidel-Morgenstern, "Relations between kinetic parameters of different column models for liquid chromatography applying core-shell particles," *Journal of Liquid Chromatography & Related Technologies*, vol. 42, no. 1-2, pp. 16–30, 2019.
- [23] G. Tingyue, T. Yung-Huoy, T. Gow-Jen, and T. George, "Modelling of gradient elution in multi-component nonlinear chromatography," *Chemical Engineering Science*, vol. 47, p. 253, 1992.
- [24] K. Miyabe and G. Guiochon, "Determination of the lumped mass transfer rate coefficient by frontal analysis," *Journal of Chromatography A*, vol. 890, no. 2, pp. 211–223, 2000.
- [25] S. Javeed, S. Qamar, A. Seidel-Morgenstern, and G. Warnecke, "Efficient and accurate numerical simulation of nonlinear chromatographic processes," *Computers & Chemical Engineering*, vol. 35, no. 11, pp. 2294–2305, 2011.
- [26] K. Kaczmarek and D. Antos, "Fast finite difference method for solving multicomponent adsorption-chromatography models," *Computers & Chemical Engineering*, vol. 20, no. 11, pp. 1271–1276, 1996.
- [27] M. d. Carmo Coimbra, C. Sereno, and A. Rodrigues, "Modelling multicomponent adsorption process by a moving

- finite element method,” *Journal of Computational and Applied Mathematics*, vol. 115, no. 1-2, pp. 169–179, 2000.
- [28] N. Rehman, M. Abid, and S. Qamar, “Numerical approximation of nonlinear and non-equilibrium model of gradient elution chromatography,” *Journal of Liquid Chromatography & Related Technologies*, vol. 44, no. 7-8, pp. 382–394, 2021.
 - [29] M. Senol, S. Atpinar, Z. Zararsiz, S. Salahshour, and A. Ahmadian, “Approximate solution of time-fractional fuzzy partial differential equations,” *Computational and Applied Mathematics*, vol. 38, no. 1, p. 18, 2019.
 - [30] G. Guiochon and B. Lin, *Modeling for Preparative Chromatography*, Academic Press, Cambridge, Massachusetts, United States, 2003.
 - [31] P. Jandera, “Can the theory of gradient liquid chromatography be useful in solving practical problems?” *Journal of Chromatography A*, vol. 1126, no. 1-2, pp. 195–218, 2006.
 - [32] B. Koren, *A Robust Upwind Discretization Method for Advection, Diffusion and Source Terms*, pp. 117–138, Centrum voor Wiskunde en Informatica, Amsterdam, 1993.
 - [33] S. Parveen, S. Qamar, and A. Seidel-Morgenstern, “Two-dimensional non-equilibrium model of liquid chromatography: analytical solutions and moment analysis,” *Chemical Engineering Science*, vol. 122, pp. 64–77, 2015.